

SUCCESSIVE DIFFERENTIATION

If $y = f(x)$ $\frac{dy}{dx}, y, y', f'(x)$

$\frac{d^2y}{dx^2}, y_2, y'', f''(x)$

In general, $\frac{dy}{dx}$ is a function of x

& its derivative w.r.t x is denoted by

$$\frac{d}{dx} \left[\frac{dy}{dx} \right] = \frac{d^2y}{dx^2}$$

Formulae:

No.	y	y_n
1.	e^{mx}	$m^n e^{mx}$
2.	a^{mx}	$m^n (\log a)^n a^{mx}$
3.	$(ax+b)^m$	$\frac{a^n m! (ax+b)^{m-n}}{(m-n)!}$ if $n < m$
		$a^n n! (ax+b)^0$ if $m=n$
4.	$\frac{1}{(ax+b)^m}$	$\frac{(-1)^n a^n (m+n-1)!}{(m-1)! (ax+b)^{m+n}}$
5.	$\frac{1}{(ax+b)}$	$\frac{(-1)^n a^n n!}{(ax+b)^{n+1}}$

$$6. \log(ax+b) \quad \frac{(-1)^{n-1} a^n (n-1)!}{(ax+b)^n}$$

$$7. \sin(ax+b) \quad a^n \sin(ax+b + \frac{n\pi}{2})$$

$$8. \cos(ax+b) \quad a^n \cos(ax+b + \frac{n\pi}{2})$$

$$9. e^{ax} \sin(bx+c) \quad x^n e^{ax} \sin(bx+c+n\theta)$$

where $x = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}(b/a)$

$$10. e^{ax} \cos(bx+c) \quad x^n e^{ax} \cos(bx+c+n\theta)$$

where $x = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}(b/a)$

① FORMULAE STEPS

$$① \quad y = e^{mx}$$

$$y_1 = m^1 e^{mx}$$

$$y_2 = m^2 e^{mx}$$

$$y_n = m^n e^{mx}$$

$$②. \quad y = \frac{1}{ax+b} \rightarrow y_1 = (-1) a \frac{1}{(ax+b)^2}$$

$$y_2 = (-1)^2 1 \cdot 2 a^2 \frac{1}{(ax+b)^3}$$

$$y_2 = (-1)^2 2! a^2 \frac{1}{(ax+b)^3}$$

$$y_n = (-1)^n n! a^n \frac{1}{(ax+b)^{n+1}}$$

3. $y = \log(ax+b)$

$$y_1 = a \cdot \frac{1}{ax+b} \quad \text{diff. (n-1) times}$$

$$a \times (-1)^{n-1} (n-1)! a^{n-1} \frac{1}{(ax+b)^n}$$

$$= (-1)^{n-1} (n-1)! a^n \frac{1}{(ax+b)^n}$$

4. $y = \frac{1}{(ax+b)^m}$ $m \in \mathbb{Z}^+$ $y_n = (-1)^n (m+n-1)! a^n \frac{1}{(ax+b)^{m+n}}$

$$y_1 = (-m) a (ax+b)^{-m-1}$$

$$= (-1) m a \frac{1}{(ax+b)^{m+1}}$$

$$= (-1) \frac{(m-1)!}{(m-1)!} \times m a \frac{1}{(ax+b)^{m+1}}$$

$$y_1 = (-1) \frac{m!}{(m-1)!} a \left(\frac{1}{(ax+b)^{m+1}} \right)$$

$$y_2 = (-1)^2 \frac{m(m+1)}{(m-0)!} a^2 \frac{1}{(ax+b)^{m+2}}$$

$$y_2 = (-1)^2 \frac{(m+1)!}{(m-0)!} a^2 \frac{1}{(ax+b)^{m+2}}$$

$$y_n = (-1)^n \frac{(m+n-1)!}{(m-0)!} a^n \frac{1}{(ax+b)^{m+n}}$$

Q Find the n^{th} derivative of $\frac{x}{(x-1)(x-2)(x-3)}$

Partial Fraction

$$\frac{x}{(x-1)(x-2)(x-3)}$$

i) Dr. is linear & repeated.

$$\frac{1}{(x+a)(x+b)} = \frac{A}{x+a} + \frac{B}{x+b}$$

(2) Dr. is linear but repeated

$$\frac{1}{(x+a)^2(x+b)} = \frac{A}{x+a} + \frac{B}{(x+a)^2} + \frac{C}{x+b}$$

3) Dr. is quadratic

$$\frac{1}{(x^2+a^2)(x^2+b^2)} = \frac{Ax+B}{x^2+a^2} + \frac{Cx+D}{x^2+b^2}$$

Note: Degree of numerator $\leq$ degree of denominator to use partial fraction.

$$\text{Let } y = \frac{x}{(x-1)(x-2)(x-3)} \rightarrow \frac{A}{(x-1)} + \frac{B}{(x-2)} + \frac{C}{(x-3)}$$

$x=1$ except $(x-1)$ bracket.

$$= \frac{1}{(1-1)(1-2)} + \frac{2}{1(1-2)(-1)} + \frac{3}{2 \cdot 1 \cdot (1-3)}$$

$$\frac{A}{(x-1)(2)} + \frac{2}{-1(x-2)} + \frac{3}{2(x-3)} \quad \left| \begin{array}{l} A = 1/2 \\ B = -2 \\ C = 3/2 \end{array} \right.$$

$$\frac{A}{(x-1)} + \frac{B}{(x-2)} + \frac{C}{(x-3)}$$

Q Find the n^{th} derivative =

① $y = \frac{4x}{(x+1)(x-1)^2}$ ② $\frac{1}{(x-2)(x-1)^2}$

③ $y = \frac{x}{(x+1)^2}$

① $\frac{4x}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$

~~$4x = A(x-1)^2$~~
 * multiply by $(x+1)(x-1)^2$ on both sides.

$$4x = A(x-1)^2 + B(x+1)(x-1) + C(x+1) = 0$$

Put $x = -1$ $-4 = A(-2)^2 \rightarrow \underline{A = -1}$

Put $x = 1$ $4 = 2C \rightarrow \underline{C = 2}$

Put $x = 0$ $0 = A - B + C$
 $B = A + C \rightarrow \underline{B = 1}$

$$y = \frac{-1}{x+1} + \frac{1}{x-1} + \frac{2}{(x-1)^2}$$

Diffⁿ y, n times w.r.t x.

$$y_n = -D^n \left(\frac{1}{x+1} \right) + D^n \left(\frac{1}{x-1} \right) + 2D^n \left(\frac{1}{(x-1)^2} \right)$$

$$D^n \left(\frac{1}{ax+b} \right) = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$D^n \left(\frac{1}{(ax+b)^m} \right) = \frac{(-1)^n (m+n-1)!}{(m-1)!} \frac{a^n}{(ax+b)^{m+n}}$$

$$y_n = - \frac{(-1)^n n!}{(x+1)^{n+1}} + \frac{(-1)^n n!}{(x-1)^{n+1}} + \frac{2(-1)^n (2+n-1)!}{(2-1)! (x-1)^{2+n}}$$

Find the n^{th} derivative.

① $y = \frac{x^3}{(x-1)(x-2)} \quad (n > 1)$

② $y = \frac{x}{x^3 - 6x^2 + 11x - 6}$

③ $y = \frac{x^4}{(x-1)(x-2)}$

calc like roots find out kya ka.

Ans

$$\begin{array}{r} x+3 \\ \hline x^3 - 3x^2 + 2x \\ \hline 3x^2 - 9x + 6 \\ \hline 7x - 6 \\ \hline \end{array}$$

$$y = Q + \frac{R}{D}$$

$$(x+3) + \frac{7x-6}{(x-1)(x-2)}$$

$x=1$ $x+3 + \frac{7(1)-6}{(1-1)(1-2)} + \frac{7(2)-6}{(2-1)(2-2)}$

$$\rightarrow (x+3) - \frac{1}{x-1} + \frac{8}{x-2}$$

Diffⁿ y, n times wrt x

$$y_n = D^n(x+3) - D^n \frac{1}{(x-1)} + 8 D^n \left(\frac{1}{x-2} \right)$$

$$y_n = 0 - \frac{(-1)^n n!}{(x-1)^{n+1}} + \frac{8(-1)^n n!}{(x-2)^{n+1}} = \frac{8(-1)^n n!}{(x-2)^{n+1}} - \frac{(-1)^n n!}{(x-1)^{n+1}}$$

$$\frac{(-1)^n n!}{(x-1)^{n+1}} \rightarrow \frac{1}{(x-1)^{n+1}}$$

$$\frac{x^4}{x^3-6x^2+11x-6}$$

$$(x-1, 2, 3)$$

★ Q. Find the n^{th} derivative of $\frac{x^B}{x^B-1}$ at $x=0$, & show that it is zero if n is even & $(-n!)$ if n is odd. ($n \geq 1$)

let $y = \frac{x^B}{x^B-1}$

No partial fraction as
Degree of numerator >
Degree of denominator.

$$\frac{x^B}{x^B-1} = \frac{x^B}{x^B-1}$$

$$\rightarrow x + \frac{x}{x^B-1}$$

$$x + \frac{1}{(x-1)(x+1)} + \frac{(-1)}{-2(x+1)} \quad (x+1) \leftarrow$$

$$y = x + \frac{1}{2(x-1)} + \frac{1}{2(x+1)}$$

diff y , n times w.r.t x .

$$y_n = 0 + \frac{1}{2} \frac{D^n}{(x-1)^{n+1}} + \frac{1}{2} \frac{D^n}{(x+1)^{n+1}}$$

Formula $D^n \frac{1}{(ax+b)} = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$

$$\frac{1}{2} \frac{d^n}{dx^n} \frac{1}{x+1} \rightarrow \frac{1}{2} \left[\frac{(-1)^n n!}{(x-1)^{n+1}} + \frac{(-1)^n n!}{(x+1)^{n+1}} \right]$$

$$\rightarrow \frac{(-1)^n n!}{2} \left[\frac{1}{(x-1)^{n+1}} + \frac{1}{(x+1)^{n+1}} \right]$$

Case-1 n is even.

$$\underline{x=0} \quad \frac{(-1)^n n!}{2} \left[\frac{1}{(-1)^{n+1}} + \frac{1}{(1)^{n+1}} \right]$$

Case-1 n is even.

$$y_n(0) = \frac{n!}{2} [-1 + 1] = \underline{\underline{0}}$$

Case-2 n is odd

$$y_n(0) = (-1) \frac{n!}{2} [1 + 1] \rightarrow \underline{\underline{-n!}}$$

★
①. $y = x \log(x+1)$ Show that $y_n = \frac{(-1)^{n-2} (n-2)! (x+n)}{(x+1)^n}$

$y = x \log(x+1)$
Diff y once w.r.t x .

$$y_1 = \frac{(x+1) \cdot 1 + \log(x+1)}{x+1} \rightarrow 1 - \frac{1}{(x+1)} + \log(x+1)$$

Diffⁿ y (n-1) times - we get.

$$y_n = 0 - D^{n-1} \left(\frac{1}{x+1} \right) + D^{n-1} \log(x+1)$$

$$= \frac{-(-1)^{n-1} (n-1)!}{(x+1)^n} + \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}}$$

$$= \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}} \left[\frac{(-1)^{n-1} (n-1)}{x+1} + 1 \right]$$

Formulae:

⑤ $D^n \sin(ax+b) = a^n \sin\left(ax+b + \frac{n\pi}{2}\right)$

⑥ $D^n \cos(ax+b) = a^n \cos\left(ax+b + \frac{n\pi}{2}\right)$

⑦ $D^n e^{ax} \sin(bx+c) = x^n e^{ax} \sin(bx+c+n\theta)$
 where $x = \sqrt{a^2+b^2}$; $\theta = \tan^{-1}\left(\frac{b}{a}\right)$

⑧ $D^n e^{ax} \cos(bx+c) = x^n e^{ax} \cos(bx+c+n\theta)$

① Find the n^{th} derivative of:

① $y = \cos x \cos 2x \cos 3x$

② $y = \cos^4 x$

③ $y = \sin^2 x \cos^3 x$

④ if $y = \sin x + \cos x$
 then $y_n = x^n [1 + (-1)^n \sin 2x]$

$$\sin(-\theta) = -\sin \theta$$

$$\cos(\theta) = \cos \theta$$

$$1 + \cos 2x = 2\cos^2 x \quad 1 - \cos 2x = 2\sin^2 x$$

classmate

Date

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Extra formulas

$$2\sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$2\cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$2\sin A \sin B = \cos(A-B) - \cos(A+B)$$

Q1

$$y = \cos x \cos 2x \cos 3x$$

$$= \frac{1}{2} (2\cos x \cos 2x) \cos 3x$$

$2\cos A \cos B$ from 2 multiply & divide on both sides of fraction line.

$$= \frac{1}{2} [\cos 3x + \cos x] \cdot \cos 3x$$

multiply $\cos 3x$ term in bracket & take 2 on both numerator as well as denominator to make it bring it in standard / assign some.

$$\frac{1}{4} [2\cos^2 3x + 2\cos x \cos 3x]$$

$$1 + \cos 2x = 2\cos^2 x$$

$$2\cos A \cos B \rightarrow \cos(A+B) + \cos(A-B)$$

$$\frac{1}{4} [1 + \cos 6x + \cos 4x + \cos 2x]$$

Diff y , n times wrt x

$$y_n = \frac{1}{4} [0 + D^n(\cos 6x) + D^n(\cos 4x) + D^n(\cos 2x)]$$

$$= \frac{1}{4} [6^n \cos(6x + \frac{2n\pi}{2}) + 4^n \cos(4x + \frac{2n\pi}{2}) + 2^n \cos(2x + \frac{2n\pi}{2})]$$

Q2

$$y = \cos^4 x$$

Step 1

$$(\cos^2 x)^2 \left(\frac{1 + \cos 2x}{2} \right)^2$$

$$(\cos^4 x) = \left(\frac{1 + \cos 2x}{2} \right)^2$$

$$y = \left(\frac{1 + \cos 2x}{2} \right)^2$$

Q3

$$y = \sin^2 x \cos^3 x$$

$$= \left(\frac{2 \sin x \cos x}{2} \right)^2 \cos x$$

bracket main multiply & divide by 2

$$= \frac{1}{4} (\sin^2 2x) \cos x$$

$\sin^2 x = \frac{1 - \cos 2x}{2}$

$$\rightarrow \frac{1}{4} \left[\frac{1 - \cos 4x}{2} \right] \cos x$$

$\sin^2 2x = \frac{1 - \cos 4x}{2}$

$$\rightarrow \frac{1}{4} \cos x - \frac{\cos 4x \cos x}{4}$$

multiply & divide by 2

$$\rightarrow \frac{1}{4} [2 \cos 5x - 2 \cos 4x \cos x]$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$\rightarrow \cos 5x + \cos(3x)$$

$$\frac{1}{16} [2 \cos x - \cos 5x + \cos 3x]$$

Diffⁿ by n times wrt x.

$$y_n = \frac{1}{16} [2^n D^n \cos x - D^n (\cos 5x) - D^n (\cos 3x)]$$

Q4

$$y = \sin 2x + \cos 2x$$

Diffⁿ y, n times wrt x

$$y_n = D^n (\sin 2x) + D^n (\cos 2x)$$

$$\rightarrow 2^n \underbrace{\sin(2x + \frac{n\pi}{2})}_{\sin \theta} + 2^n \underbrace{\cos(2x + \frac{n\pi}{2})}_{\cos \theta}$$

$$2^n [\sin \theta + \cos \theta]$$

$$\rightarrow 2^n \left[\sqrt{\sin^2(2x + \frac{n\pi}{2}) + \cos^2(2x + \frac{n\pi}{2})} \right]^2$$

$$\rightarrow 2^n \sqrt{1 + 2 \sin(2x + \frac{n\pi}{2}) \cos(2x + \frac{n\pi}{2})}$$

$$\rightarrow 2^n \sqrt{1 + \sin 2(2x + \frac{n\pi}{2})}$$

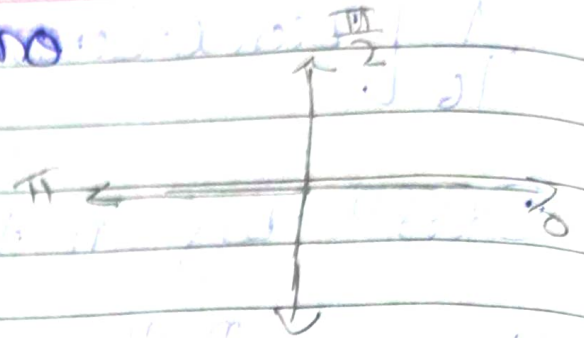
$$\rightarrow 2^n \sqrt{1 + \sin(4x + n\pi)}$$

$$\sin(n\pi + \theta) = (-1)^n \sin \theta$$

$$\sin(\pi + \theta) = -\sin \theta$$

$$\sin(2\pi + \theta) = \sin \theta$$

$$\sin(n\pi + \theta) = (-1)^n \sin \theta$$



$$x^n + [1 + (-1)^n \sin 2\theta x]^{1/2}$$

hence Proved.

~~$$y = 2^{2x} \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$$~~

~~$$y = 2^{2x} \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$$~~

~~$$y = 2^{2x} \sin^2 x \cos^2 x$$~~

~~$$y = 2^{2x} \cos x \cos 3x$$~~

~~$$x = e^{\log x}$$~~

$$x = e^{\log x}$$

$$2^x = e^{\log(2^x)}$$

$$= e^{(\log 2)x}$$

$$\Rightarrow y = e^{2x} \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$$

$$\sin x = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$$

$$y = \frac{1}{2} e^{2x} (2 \sin x/2 \cos x/2)$$

$$= \frac{1}{2} e^{2x} \sin x$$

$$D^n [e^{ax} \sin(bx+c)] = x^n e^{ax} \sin(bx+c+n\theta)$$

$$y_n = \frac{1}{2} D^n [e^{2x} \sin x] \quad \underline{a=2, b=1}$$

$$y_n = \frac{1}{2} x^n e^{2x} \sin(x+n\theta)$$

$$\text{where } x = \sqrt{a^2 + b^2} = \sqrt{5}$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{1}{2}\right)$$

$$y_n = \frac{1}{2} (5)^{n/2} e^{2x} \sin(x + n \tan^{-1} 1/2)$$

Find nth derivative of

1) $y = \tan^{-1} x$

2) $y = \tan^{-1} \left(\frac{1+x}{1-x} \right)$

3) $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$

4) $y = \cos^{-1} \left(\frac{x - 1/x}{x + 1/x} \right)$

$$\sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$$

$$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

$$\tan x = \frac{2 \tan x}{1 - \tan^2 x}$$

2 $y = \tan^{-1} \left(\frac{1+x}{1-x} \right)$

Put $x = \tan \theta \rightarrow \theta = \tan^{-1} x$

$\frac{1-x}{1+x} = \tan \frac{\pi}{4}$

$$y = \tan^{-1} \left[\frac{\tan \frac{\pi}{4} + \tan \theta}{1 - \tan \frac{\pi}{4} \tan \theta} \right]$$

$$\tan^{-1} \left[\tan \left(\frac{\pi}{4} + \theta \right) \right]$$

$\frac{\pi}{4} + \theta \rightarrow \frac{\pi}{4} + \tan^{-1} x = y$

3) $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$ $x = \tan \theta$

$$\sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right)$$

4) $y = \cos^{-1} \left(\frac{x-1/x}{x+1/x} \right)$ $x = \tan \theta$

$$\cos^{-1} \left[\frac{x^2 - 1}{x^2 + 1} \right] \rightarrow \cos^{-1} \left[-\frac{(1-x^2)}{1+x^2} \right]$$

$$\cos^{-1}(-\cos 2\theta) = x = \tan \theta \rightarrow y = \cos^{-1}[-\cos 2\theta]$$

$$= \cos^{-1}[\cos(\pi + 2\theta)]$$

$$x^2 + a^2 = (x - ia)(x + ia) \quad x^2 - a^2 = (x - a)(x + a)$$

$$y_1 = \frac{1}{x^2 + 1} = \frac{1}{(x - i)(x + i)}$$

$$y_1 = \frac{1}{(x - i)(i + i)} + \frac{1}{(-i - i)(x + i)} \rightarrow \frac{1}{2i(x - i)} - \frac{1}{2i(x + i)}$$

Diff y , $(n-1)$ times w.r.t x

$$D^n \left(\frac{1}{ax + b} \right) = \frac{(-1)^n n! a^n}{(ax + b)^{n+1}}$$

$$D^{n-1} \left(\frac{1}{ax + b} \right) = \frac{(-1)^{n-1} (n-1)! a^{n-1}}{(ax + b)^n}$$

$$y_1 = \frac{1}{2i} \frac{D^n (x - i)^{-1}}{(x - i)^{n+1}} - \frac{1}{2i} \frac{D^n (x + i)^{-1}}{(x + i)^{n+1}}$$

$$y_n = \frac{1}{2i} D^{n-1} \left(\frac{1}{x-i} \right) - \frac{1}{2i} D^{n-1} \left(\frac{1}{x+i} \right)$$

$$y_n = \frac{1}{2i} \frac{(-1)^{n-1} (n-1)!}{(x-i)^n} - \frac{1}{2i} \frac{(-1)^{n-1} (n-1)!}{(x+i)^n}$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{2i} \left[\frac{1}{x-i} - \frac{1}{x+i} \right]$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{2i x^n} \left[\frac{1}{\cos \theta - i \sin \theta} - \frac{1}{\cos \theta + i \sin \theta} \right]$$

$$\frac{(-1)^{n-1} (n-1)!}{2i x^n} \left[(\cos \theta + i \sin \theta)^n - (\cos \theta - i \sin \theta)^n \right]$$

$$y_n = (-1)^{n-1} (n-1)! \sin^n \theta \sinh \theta.$$

LEIBNITZ'S THEOREM

If u and v are two differentiable functions in x ,

then

$$(uv)^n = n(u^{n-1}v) + (u^n)v$$

If u & v are two then $(uv)^n = (u^n)v + n(u^{n-1}v)$

Find n^{th} derivative of

① $y = x^3 e^{mx}$

② $y = x \log(x+1)$

③ $y = x^n \log x$, ST
 $y_{nm} = \frac{n!}{x}$

① $y = x^3 e^{mx} \rightarrow e^{mx} x^3$

$U_n = m^n e^{mx}$

$U_{n-1} = m^{n-1} e^{mx}$

$V_1 = 3x^2$
 $V_2 = 6x$
 $V_3 = 6$

By Leibnitz's theorem.

$(UV)_n = (U_n V) + nC_1 U_{n-1} V' + \dots + nC_n U V^{(n)}$

$y_n = m^n e^{mx} x^3 + n m^{n-1} e^{mx} (3x^2) + \frac{n(n-1)}{2} m^{n-2} e^{mx} (6x)$

$= e^{mx} \left[m^n x^3 + 3n m^{n-1} x^2 + 3n(n-1) m^{n-2} x \right]$

2 $y = x \log(x+1) = \log(x+1) \cdot x$

$u = \log(x+1)$ $v = x$

$D^n \log(ax+b) = \frac{(-1)^{n-1} (n-1)!}{(ax+b)^n}$

~~$U_n = \frac{(-1)^{n-1} (n-1)!}{(x+1)^n}$~~

~~$V = x$~~

~~U_{n-1}~~

$$U_n = \frac{(-1)^{n-1} (n-1)!}{(x+1)^n}$$

$$U_{n-1} = \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}}$$

$$V=1$$

By LT

$$(UV)_n = (U_n)V + n^4 (U_{n-1})V_1 + \dots$$

$$y_n = \frac{(-1)^{n-1} (n-1)!}{(x+1)^n} \cdot x + \frac{n(-1)^{n-2} (n-2)!}{(x+1)^{n-1}} \quad (1)$$

$$= \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}} \left[-\frac{(n-1)x}{x+1} + n \right]$$

$$= \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}} \left[-\cancel{nx} + x + nx + n \right]$$

$$= \frac{(-1)^{n-2} (n-2)!}{(x+1)^{n-1}} \frac{(x+n)}{x+1}$$

3 $y = x^n \log x$

Diffⁿ y, once wrt x.

$$y_1 = \frac{x^n}{x} + (\log x) n x^{n-1}$$

multiply by x throughout.

x^n by e

$$xy_1 = x^n + n(\log x)x^n \Rightarrow y_1 x = x^n + ny \quad \text{--- (1)}$$

Diffⁿ (1) n times w.r.t x

$$(y_1 x)^n = (x^n)^n + (ny)^n \quad \text{--- (2)}$$

$$(y_1 x)^n = (y_1)^n x + n(y_1)^{n-1} (1) (\log x)$$

$$= y_{n+1} x + n y_n \quad \text{--- (3)}$$

$$(x^n)^n = n! \quad \text{--- (ii)}$$

$$(ny)^n = n y_n \quad \text{--- (iii)}$$

Sub (i), (ii), (iii) $y_{n+1} x + n y_n = n! + n y_n$

$$y_{n+1} = \frac{n!}{x}$$

① $y = \cos^{-1} x$ ST $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0$

② $y = (x + \sqrt{a^2 + x^2})^n$ ST, $x=0$ $a^2 y_{n+2} + (n^2 - m^2)y_n = 0$

③ $y = e^{m \cos^{-1} x}$ ST

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + m^2)y_n = 0$$

4

$$y^{1/m} + y^{-1/m} = 2x \quad \text{ST,}$$

$$(x^2-1)y_{n+2} + (2n+1)x y_{n+1} + (n^2-m^2)y_n = 0.$$

Formula to be used:

$$(uv)_n = (u_n)v + nC_1(u_{n-1})v_1 + nC_2(u_{n-2})v_2 + \dots + nC_n(u_1)v_n$$

$$\textcircled{1} \quad y = \cos^{-1} x \quad \textcircled{1}$$

Diff $\textcircled{1}$ wrt x

$$y_1 = \frac{-1}{\sqrt{1-x^2}} \rightarrow y_1 \sqrt{1-x^2} = -1 \quad \textcircled{2}$$

Diff $\textcircled{2}$ wrt x

$$y_2 \sqrt{1-x^2} + y_1 \frac{(-2x)}{2\sqrt{1-x^2}} = 0.$$

$$y_2(1-x^2) - y_1 x = 0 \quad \textcircled{3}$$

Diff $\textcircled{3}$, n times wrt x , using LT.

$$(uv)_n = (u_n)v + nC_1(u_{n-1})v_1 + nC_2(u_{n-2})v_2 + \dots +$$

$$[y_2(1-x^2)]_n - [y_1 x]_n = 0.$$

$$(y_2)_n (1-x^2)_n - (y_1)_n x_n = 0$$

$$\left[y_{n+2}(1-x^2) + n y_{n+1}(-2x) + \frac{n(n-1)}{2} y_n(-2) \right] = 0$$

First order ka derivation function
 2nd term constant remains last term

$$\left[y_{n+1}(1) + n y_n(1) \right] = 0$$

$$(1-x^2)y_{n+2} + () y_{n+1} + () y_n = 0$$

1. find x to x comparison @ 6610

2. $y = (x + \sqrt{a^2 + x^2})^n$

$$0 = n y^{n-1} \cdot \frac{d}{dx} (x + \sqrt{a^2 + x^2}) + n \left[(-x + \frac{x}{\sqrt{a^2 + x^2}}) y^n \right]$$

Diff ① wrt x

$$y_1 = n(x + \sqrt{a^2 + x^2})^{n-1} \left(1 + \frac{x}{\sqrt{a^2 + x^2}} \right)$$

$$\left[n y^{n-1} (x + \sqrt{a^2 + x^2})^{n-1} \left(\frac{\sqrt{a^2 + x^2} + x}{\sqrt{a^2 + x^2}} \right) + n y^n \left(-x + \frac{x}{\sqrt{a^2 + x^2}} \right) \right] = 0$$

$$y_1 \sqrt{a^2 + x^2} = n(x + \sqrt{a^2 + x^2})^n = n y \quad \text{--- (1) (from eqn ①)}$$

$$y_1 \sqrt{a^2 + x^2} - n y = 0 \quad \text{--- (2)}$$

Diff ② wrt x by (B) L. C. method

$$y_2 \sqrt{a^2 + x^2} + y_1 \frac{2x}{2\sqrt{a^2 + x^2}} - m y_1 = 0$$

$$y_2 (a^2 + x^2) + y_1 x - m y_1 \sqrt{a^2 + x^2} = 0$$

$$y_2 (a^2 + x^2) + y_1 x - m(m y_1) = 0 \quad \text{--- (3)}$$

from eqn (2)

Diff ③ n times wrt x by LT.

$$[y_2 (a^2 + x^2)]_n + [y_1 x]_n - m^2 y_n = 0$$

$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ u & v & u & v & u & v \end{matrix}$

$$(uv)_n \rightarrow (un)_v + {}^n C_1 u_{n-1} v + \dots$$

$$+ \left[(n y_{n+2}) (a^2 + x^2) + n y_{n+1} 2x + n(n-1) y_{n-2} \right] + \left[n y_{n+1} x + n y_n (1) \right] - m^2 y_n$$

$\begin{matrix} (n y_{n+2}) (a^2 + x^2) \\ (n y_{n+1} 2x) \\ (n y_n (1)) \end{matrix}$

$$\text{--- (4) --- } 0 = 6m - 5m = m$$

$$(a^2 + x^2)y_{n+2} + (2n+1)x y_{n+1} + (n^2 - m^2)y_n = 0 \quad \text{--- (4)}$$

Put $x=0$ in (4).

$$a^2 y_{n+2}(0) + 0 + (n^2 - m^2)y_n(0) = 0.$$

$$a^2 y_{n+2}(0) + (n^2 - m^2)y_n(0) = 0$$

EXPANSION OF FUNCTIONS

① Taylor's Series about $x=a$

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots \quad \text{--- (1)}$$

② Taylor's Series in power of x .

$$f(x+h) = f(h) + \frac{f'(h)}{1!}x + \frac{f''(h)}{2!}x^2 + \dots \quad \text{--- (2)}$$

③ Taylor's Series about origin ($x=0$)
(Maclaurin's series).

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots \quad \text{--- (3)}$$

① Expand e^x in powers of $(x-1)$
or
Expand e^x using Taylor's Series about $x=a$ is

Formula: $f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$

put $a=1$ $f(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \dots$ ②

$f(x) = e^x$	$f(1) = e$
$f'(x) = e^x$	$f'(1) = e$
$f''(x) = e^x$	$f''(1) = e$

Subs in ② $e^x = e + e(x-1) + \frac{e}{2!}(x-1)^2 + \dots$

* ② Expand $\tan^{-1}x$ in power of $(x-\pi/4)$

Questions

① Expand $\sin x$ in power of $x+\pi/4$

② Expand $\log x$ in power of $(x-1)$ & hence ST
 $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

HW

* ③ Expand $x^3 - 14x^2 + x - 1$ in power of $(x-1)$

Ans ① $x + \frac{\pi}{4} \rightarrow x - (-\frac{\pi}{4})$

Taylor Series

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

put $a = -\frac{\pi}{4}$

$$f(x) = f\left(-\frac{\pi}{4}\right) + \frac{f'\left(-\frac{\pi}{4}\right)}{1!}\left(x + \frac{\pi}{4}\right) + \frac{f''\left(-\frac{\pi}{4}\right)}{2!}\left(x + \frac{\pi}{4}\right)^2 + \dots \text{--- ②}$$

$$f(x) = \sin x \quad \left| \quad f\left(-\frac{\pi}{4}\right) = \sin\left(-\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} \right.$$

$$f'(x) = \cos x$$

$$f'\left(-\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$f''(x) = -\sin x$$

$$f''\left(-\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$f'''(x) = -\cos x$$

$$f'''\left(-\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$f^{(4)}(x) = \sin x$$

$$f^{(4)}\left(-\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

Final Series

$$\sin x = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\left(x + \frac{\pi}{4}\right) - \frac{1}{\sqrt{2}}\frac{\left(x + \frac{\pi}{4}\right)^2}{2!} + \frac{1}{\sqrt{2}}\frac{\left(x + \frac{\pi}{4}\right)^3}{3!} - \frac{1}{\sqrt{2}}\frac{\left(x + \frac{\pi}{4}\right)^4}{4!} + \dots$$

3. $\log x$ in power of $(x-1)$

sr $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

By TS $f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$

$f(x) = \log x$ $f(1) = 0$

$f'(x) = \frac{1}{x}$ $f'(1) = 1$

$f''(x) = -\frac{1}{x^2}$ $f''(1) = -1$

$f'''(x) = \frac{2}{x^3}$ $f'''(1) = 2$

$f^{(4)}(x) = -\frac{6}{x^4}$ $f^{(4)}(1) = -6$

$\log x = 0 + \frac{1}{1!}(x-1) - \frac{1}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 - \frac{6}{4!}(x-1)^4 + \dots$

$1(x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + \frac{1}{4}(x-1)^4$

~~$\log x = (x-1) \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right]$~~

Approximation problems no need.

classmate

Date

Page

$$\log x = (x-1) \left[1 - \frac{1}{2}(x-1) + \frac{1}{3}(x-1)^2 - \frac{1}{4}(x-1)^3 + \dots \right]$$

$$\underline{x=2}$$

$$\log 2 = 1 \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right]$$

hence proved

2nd formula ques

$$\boxed{f(x+h) = f(h) + \frac{f'(h)}{1!}x + \frac{f''(h)}{2!}x^2 + \dots}$$

① Using TS, arrange in powers of x

$$7 + (x+2) + 3(x+2)^3 + (x+2)^4 - (x+2)^5$$

② Find the expansion of $\tan(x+\pi/4)$ in powers of x , upto the term x^4 .

Ans ① $\therefore f(x+h) = f(h) + \frac{f'(h)}{1!}x + \frac{f''(h)}{2!}x^2 + \dots$

Put $h=2$

$$f(x+2) = f(2) + \frac{f'(2)}{1!}x + \frac{f''(2)}{2!}x^2 + \frac{f'''(2)}{3!}x^3 + \dots$$

where $f(x+2) = 7 + (x+2) + 3(x+2)^3 + (x+2)^4 - (x+2)^5$

$$\therefore f(x) = 7 + x + 3x^3 + x^4 - x^5$$

$$f'(2) = 17$$

$$f(x) = 1 + 9x^2 + 4x^3 - 9x^4$$

$$f''(x) = 18x + 12x^2 - 20x^3$$

$$f'''(x) = 18 + 24x - 60x^2$$

$$f^{(4)}(x) = 24 - 120x \quad f^{(5)}(x) = -120$$

$$f'(2) = -11 \quad f''(2) = 76 \quad f'''(2) = -174$$

$$f^{(4)}(2) = -216$$

Subs in ① -

$$f(x+2) = 17 - 11x - \frac{76}{2}x^2 - \frac{174}{6}x^3 - \frac{216}{24}x^4 - \frac{120}{120}x^5$$

$$f(x+2) = 17 - 11x - 38x^2 - 29x^3 - 9x^4 - x^5$$

2. Find the expansion of $\tan(x + \frac{\pi}{4})$ in powers of x , upto term x^4 .

$$\begin{aligned} \text{IS } f(x+h) &= f(h) + \frac{f'(h)}{1!}x + \frac{f''(h)}{2!}x^2 \\ &+ \frac{f'''(h)}{3!}x^3 + \frac{f^{(4)}(h)}{4!}x^4 + \dots \end{aligned}$$

Given $f(x + \frac{\pi}{4}) = \tan(x + \frac{\pi}{4})$ — ①

$$f(x) = \tan x$$

$$f(x) = \tan x$$

$$f(\pi/4) = 1$$

$$f'(x) = \sec^2 x$$

$$f'(\pi/4) = 1 + f^2(\pi/4) = 2$$

$$f''(x) = 1 + \tan^2 x$$

$$f''(x) = 1 + f^2(x)$$

$$f''(x) = 2 f(x) f'(x)$$

$$f''(\pi/4) = 2(1)(2) = 4$$

$$f'''(x) = 2[f(x)f''(x) + f'(x)f''(x)]$$

$$f'''(\pi/4) = 2[(1)(4) + (2)(4)]$$

$$+ 2f(x)f'(x)]$$

$$= 16$$

$$f^{(4)}(x) = 2[f(x)f'''(x) + f'(x)f''(x) + 2f(x)f''(x)]$$

$$f^{(4)}(\pi/4) = 2[(1)(16) + (2)(4) + 2(2)(4)]$$

$$= 2[16 + 8 + 16]$$

$$= 80$$

$$\frac{168}{3 \times 2}$$

$$\frac{80}{10}$$

$$\frac{80}{4 \times 3 \times 2}$$

$$f(x+h) = 1 + \frac{2x}{1!} + \frac{4}{2!}x^2 + \frac{16}{3!}x^3 + \frac{80}{4!}x^4$$

$$1 + 2x + 2x^2 + \frac{8}{3}x^3 + \frac{10}{3}x^4$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \frac{2x}{1} = 2x$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h^2} = \frac{2x}{2} = x$$